

Markscheme

Mathematics: analysis and approaches
Practice question bank

63. (a)

The first term of an arithmetic sequence is 8 and the 15th term is 71.

Find the common difference, and hence find the sum of the first 15 terms.

$$u_{15} = u_1 + 14d: 71 = 8 + 14d \quad (M1)$$

$$d = 4.5 \quad \mathbf{A1}$$

$$S_{15} = \frac{15}{2}(u_1 + u_{15}) = \frac{15}{2}(8 + 71) \quad (M1)$$

$$S_{15} = 592.5 \quad \mathbf{A1}$$

[5 marks]